

Evaluating Cluster Dendrograms for Spatiotemporal Anomaly Tracking

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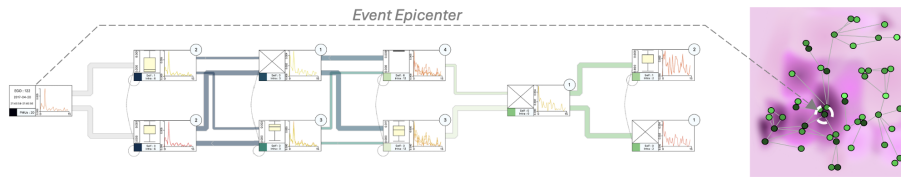


Figure 1: An example of a cluster dendrogram (LHS) for a spatiotemporal anomaly event, generated for a voltage fluctuation event over a synthetic power grid network with 50 smart sensors (RHS). Each dendrogram column (left to right) represents a topologically connected ‘hop’ level from the event epicenter, with cluster nodes showing averaged sensor data streams, sensor counts, and correlation to the epicenter.

Abstract

Spatiotemporal network analyses face growing challenges due to the rising deployment of ‘smart’ sensors across domains such as power, seismology, and transportation, where high interconnectivity and large volumes of noisy data complicate efficient monitoring. Proposed visualization solutions have largely focused on custom glyphs or interactive lenses in node-link diagrams, with spatiotemporal clustering used to mitigate scale. Recently, ‘cluster dendrograms’ have been proposed to summarize how anomalous events propagate from an epicenter through a network; however they abstract topology, and have not been empirically evaluated against topology-preserving strategies. We contribute a controlled user study evaluating efficiency, effectiveness, and confidence in anomaly tracking tasks using sensor data from both power transmission and seismic networks. Our results indicate that cluster dendrogram layouts support users better than node-link diagrams, especially when clutter is present.

CCS Concepts

• **Human-centered computing** → *Dendrograms; Empirical studies in visualization; Visualization design and evaluation methods*; • **Computer systems organization** → *Sensor networks*; • **Networks** → *Network monitoring*;

1. Introduction

Identifying and tracking anomalous events is an important task in many spatiotemporal analyses. For example, domains such as power, seismology, transportation, agriculture, IoT, etc. deploy ‘smart’ sensors in spatiotemporal networks, which enable real-time wide-area monitoring and control [SYPB21]. For event identification and tracking, both confirmatory and exploratory methods have been developed [LDR20, ELL⁺11]. However, the high interconnectivity of sensors in these domains can make it challenging to isolate specific network events, and the large, concurrent volumes of noisy data that is generated can make it difficult to quickly identify and track any anomalous patterns or behaviors [APSB23, AA10]. Traditionally, node-link diagrams superimposed onto geographic maps can be manually cross referenced with charts showing data streams for each node [APSB23]. Sev-

eral platforms have extended this technique to encode additional information, such as overlaying a contour map to represent attribute profiles [GOSW16, She17, GMJ23], using custom glyphs to reflect node characteristics [OW00, JK23], or using interactive lenses to selectively reveal information [BKA⁺16, WC11].

Alternatively, several recent approaches in spatiotemporal data science have advocated for space-time clustering approaches that leverage agglomerative hierarchical clustering to identify groupings in network data [SJA⁺15, MBMM12, KH03, PDBKS10], and consequently reduce the network scale for analyses. This complements the development of a recently-introduced visualization technique named *cluster dendrograms*, originally designed to support *epicentric analysis* of anomalous events in a power grid network (see Fig. 1 for an example image) [APSB23]. Epicentric refers to events that originate from some epicenter/source location and prop-

agate outward through the network topology [APSB23, KSU*14, HDLT05]. Cluster dendrograms were designed to support such analysis, in part by abstracting the grid network topology. However, while the technique is in theory domain-agnostic for tasks relating to epicentric analysis of events that occur over a spatiotemporal network, it has thus far not been (i) experimentally evaluated nor (ii) applied to anomaly tracking tasks in different domains.

In this work, we empirically evaluate cluster dendrograms as a general-purpose visualization technique via a controlled experiment using both power grid data [LYBO21] and seismic wave data [FBTM*19]. Both domains involve anomaly events that occur over space and time (though they are caused by widely different phenomenon) stemming from an identifiable epicenter, and involve large-scale networks of sensors that generate continuous streams of data, requiring efficient methods for identifying and tracking anomalous events amidst significant noise.

To perform our study, we compared the cluster dendrogram technique against ‘baseline’ displays suitable for comparison (based on node-link diagrams). Subjects were shown several instances of network layouts and events in a within subjects protocol, and tasked to perform epicentric analysis operations. We collected participant task accuracy, response time, and confidence ratings. In subsequent analysis, we found that decisions with cluster dendrograms were (i) better on average, having higher accuracy (0.87 ± 0.08 vs. 0.54 ± 0.11), lower response time (8.66 ± 0.86 vs. 19.56 ± 1.09 seconds), and higher confidence (4.53 ± 0.11 vs. 3.08 ± 0.18 out of 5); and (ii) such performance was consistent for both domain experts (i.e., participants familiar with the underlying power or seismic domain) vs. non-experts (participants unfamiliar with the underlying domain). We also found that the node-link diagram based interfaces were more sensitive to scale, and participants were less able to determine the direction that events propagated. Based on our study results, we discuss implications for the cluster dendrogram technique for real-time spatiotemporal event tracking and extensibility to domains involving dynamic networks.

2. Stimuli Design

Sensor Data: To generate stimuli from the power and seismic datasets used in our study (described in Sec. 3.1), we calculated Fast Fourier transforms (FFTs) on the sensor data [MVRE06, AGP*22, CBB19], which is commonly done to characterize the time evolution of the ‘anomaly events’, i.e., anomalous high frequency/magnitude fluctuations in sensor measurements (for power systems this includes voltage fluctuation magnitudes, and for seismic data this includes P-wave magnitudes, which are typically the first waves detected by seismometers when an earthquake occurs).

Network Links: For power system networks, if there exists a transmission line connecting two sensors, then this constitutes a link in the node-link diagram. Seismic sensor networks are not connected by physical lines; however, their location and/or measurements can be leveraged to create a network for analysis purposes [HA04]. For seismic data, links were thus created for sensors geographically located within 5km from each other. We constructed node-link diagrams using force directed layout to show the full sensor network (Fig.2(A)) based on the above processing. A time slider

was present to cover a two hour period centered on the anomaly event’s occurrence. Here, each node corresponded to a line chart of its raw FFT data stream calculated for a 10 second period starting from the selected time stamp on the time slider. The darkness of the FFT line corresponded to the similarity the node exhibited to the epicenter. We opted for force-directed layout to reflect common operational practice in power and seismic monitoring [PM09, HA04], (as a note: a geographically-grounded layout may also be employed here, but we leave this as a direction for future work).

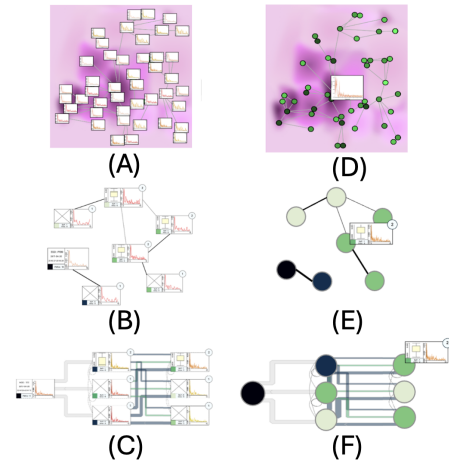


Figure 2: Stimuli settings: a default cluster dendrogram (C), a standard node-link diagram (D), and four variations (A, B, E, F). These comprise a sensor node-link diagram with overlaid FFT plots (A) or FFT tooltip (D); a clustered node-link diagram with box-glyphs (B) or box-glyph tooltip (E); and a clustered dendrogram with box-glyphs (C) or box-glyph tooltip (F). All views include a time slider spanning a 2-hour window centered on the anomalous events.

Clustering Network Sensors: In alignment with the cluster dendrogram technique [APSB23], we performed agglomerative hierarchical clustering [LWS18] across the network with respect to a pre-identified epicenter sensor, considering the FFT-similarity and network hop-distance for each sensor relative to the epicenter. Similarity thresholds (see Supplemental Material) were selected to finalize clustering granularity. We constructed stimuli showing the clustered sensor network using a force-directed node-link diagram (Fig.2(B)). Cluster dendrograms represent cluster nodes as a box glyph where: (i) a color swatch (bottom-left) indicates correlation with the epicenter (darker $\equiv$ higher correlation); (ii) a line chart of the average FFT data for the cluster’s sensors; (iii) a circle icon (top-right) indicating the number of sensors whose peak magnitude reached at least 70% of the epicenter’s; and (iv) a box plot of the distribution of peak FFT magnitudes. A link is drawn between clusters when there existed at least one connection between the constituent sensors across clusters; link thickness encodes FFT-Similarity averaged across clusters. Additionally, self-links represent connections between sensors within the same cluster, depicted as circular arcs in the bottom-left corner of each node box. Mouseover of clustered network nodes renders a tooltip listing the IDs of the constituent sensor nodes.

Cluster Dendrogram: The ‘default’ cluster dendrogram tech-

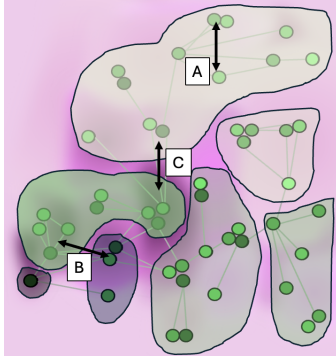


Figure 3: (A) Self links connect nodes within a cluster. (B) Intra-hop links connect nodes in different clusters at the same hop distance from the epicenter. (C) Inter-hop links connect nodes in different clusters at different hop distances from the epicenter.

nique (Fig.2(C)) visualizes the final clustering result, such that each level of the dendrogram reflected increasing network hop-distance from the epicenter sensor; consequently there may be multiple clusters located at the same dendrogram level. Clusters are ordered within each dendrogram level such that the most similar cluster of that level was located at the top. As is common in power and seismic domains, there is a congruence between increasing hop distance and increasing cluster dissimilarity in the dendrogram (a consequence of the event propagation mechanic exhibited by the power and seismic sensor networks). Nodes were styled as in Fig.2(B); however, we represented three types of links (Fig.3): (i) Self-links (sensors within-cluster) as circular arcs in the bottom-left corner of the node; (ii) Intra-hop (sensors at the same hop level, different clusters) as dotted and arced lines between the clusters; and (iii) Inter-hop (sensors in different hops) as colored flows (based on cluster or origin) whose thickness encodes downstream influence calculated based on the relative Manhattan distances of the cluster-wise averaged FFT data.

In addition to the ‘default’ dendrogram node boxes, we also tested equivalent versions of stimuli where the nodes are represented as circles (Figs. 2(D–F)). Circle color indicated the similarity of sensor/cluster to that of the epicenter. The FFT data streams were visible on mouseover of the sensor/cluster node.

3. Empirical Evaluation

3.1. Study Design and Procedure

Stimuli Data. For our study, we generated a set of 36 data snapshots of anomaly events using 2 datasets: (i) 18 power sensor snapshots measuring voltage data pertaining to an oscillation event from our power system dataset [LYBO21], and (ii) 18 snapshots of P-wave seismic sensor data during earthquakes recorded at 1Hz over a 1 minute period [FBTM*19]. All the data snapshots used a network of 50 sensors with varying network shapes; all the network shapes have a similar degree distribution and network density, and can therefore be considered equivalent [FLNU18]. Additionally, the anomaly event data used exhibits similar event duration and

epicenter magnitude. Each stimuli setting (Fig. 2) was associated with 3 data snapshots each for power and seismic data respectively.

Study Design. We summarize the study procedure in Fig. 4. Trials were partitioned into two blocks based on the dataset (power/seismic) used; block order was randomized. Within each block, participants had to perform trials across all 6 stimuli settings in a randomized order. In each stimuli setting, participants had to complete the following tasks, designed to mimic real-world epicentric event analysis operations in power and seismic monitoring:

- (ST1) Identify which sensor in a given cluster behaves most similarly to another cluster (point-to-point comparison).
- (ST2) Identify which neighboring cluster maximally contributes to a given cluster (group-level comparison).
- (ST3) Determine if a given cluster is strongly or weakly influenced by another specified cluster not directly connected by an edge (propagation tracking across non-adjacent clusters).

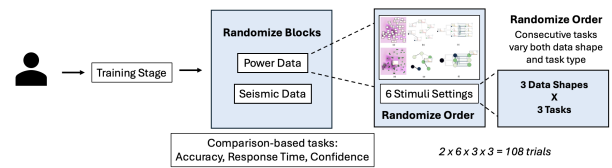


Figure 4: Study design as described below. Participants complete 54 trials per data domain. Post study, participants also rank the stimuli settings in order of preference.

Each task was completed 3 times within a stimuli setting (one corresponding to each of the 3 data snapshots); the task order was randomized such that no participant performed the same task twice in a row, nor did they encounter the same data snapshot in consecutive trials. Therefore, in total participants completed 108 trials (2 datasets x 6 stimuli settings x 3 data snapshots x 3 tasks). For each trial, the collected data was: response correctness, response time, and the participant’s self-report of confidence (Low, Somewhat Low, Neutral, Somewhat High, High). Prior to starting the main study, 18 training trials (1 per unique task + stimuli setting combo) were run for the given interface from a set of 6 additional data snapshots that were reserved for training trials (3 each from power and seismic domains). After completing the main study, participants were then asked to rank which interface they were most comfortable using, and completed a demographic survey.

Participants and Apparatus. A pilot study was conducted with 3 participants to validate the study design and data snapshots. Then, 45 participants completed the study (18 males, 12 females, average age=27.03, SD=3.27). All participants were graduate students at the XYZ University. Using a 5-point Likert scale, participants also self-reported their familiarity with data visualization (average=2.60, SD=0.62). We also intentionally recruited participants (see Supplemental material for demographic details) such that: 15 participants were experts in power data (i.e., expertise ≥ 3), 15 in seismic data, and 15 with no expertise in either domain. This allowed us to assess the feasibility of the technique both for domain ‘experts’ and ‘non-experts.’ Study sessions were conducted in a quiet, office-like environment with no distractions, with a 24-inch desktop monitor using Google Chrome (1920 x 1080 resolution).

4. Analysis and Discussion

4.1. Study Results

We performed iterations of factor analysis [Hai09, Kef10] to determine the effect strength of accuracy, response time, and confidence (see Supplemental Material). ST2 failed to load for the cluster dendrogram settings Fig. 2(C,F), for both datasets across all the measures, while ST1 and ST3 failed to load for response time. Hence we performed t-tests for each performance variable individually as a dependent variable, across each task type x stimuli setting (independent variable). All results were statistically significant ($p < 0.05$). We address the following hypotheses:

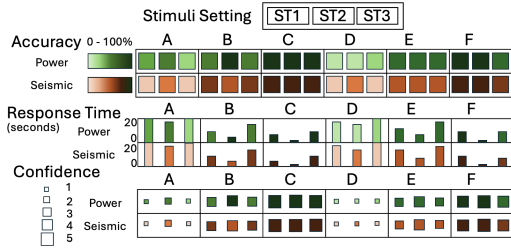


Figure 5: Performance accuracy, response time, and confidence (mapped to a 5-point Likert scale) across all 6 stimuli settings. Cluster dendrograms (C), (F) show the greatest improvement.

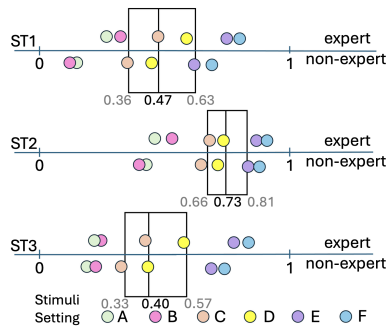


Figure 6: Accuracy distribution of Expert vs. Non-Expert performance across stimuli settings; average of power and seismic data.

H1: Cluster dendrograms outperform node-link diagrams across all measures and tasks. This hypothesis is fully supported. We observed that for the stimuli settings with cluster dendrograms (Fig. 2 (C,F)), participants exhibited an overall increase in both accuracy (27.3% $\uparrow$; $\bar{\beta} = 0.73$) and confidence (31.2% $\uparrow$, $\bar{\beta} = 0.68$), and decrease in response time (52.4% $\downarrow$, $\bar{\beta} = 0.77$) across all three tasks, in comparison to the clustered node-link ($\bar{\beta}_{Acc} = 0.63$; $\bar{\beta}_{RT} = 0.67$; $\bar{\beta}_{Conf} = 0.65$) and sensor node-link diagrams ($\bar{\beta}_{Acc} = 0.52$; $\bar{\beta}_{RT} = 0.43$; $\bar{\beta}_{Conf} = 0.49$). We summarize our results in Fig. 5. We also observed that (C) is the highest performer, indicating that cluster dendrograms are capable of supporting high dimensional data. In general, ST3 showed the greatest improvement across response time, and ST2 for accuracy and confidence. Also, seismic data gains were marginally greater than power data across all tasks.

H2: Non-experts perform worse than experts for node-link diagrams, and perform comparably with experts for cluster

dendrograms. This hypothesis is partially supported. We find that all 3 participant groups considered ST2 to be ‘trivial’ with cluster dendrograms ($\bar{\beta}_{Acc} = 0.95$; Cohen’s $d=0.78$). The effect sizes for ST1 ($d=0.64$), ST3 ($d=0.55$) were lower, with non-expert participants performing with slightly less efficiency (16.3% $\downarrow$ in accuracy) than those who had expertise in at least one domain. The effect sizes were moderate for clustered node link diagrams across all tasks ($\bar{d}=0.48$), and low ($\bar{d} < 0.3$) when the sensor network was shown. Non-experts also performed slightly better (12.5% $\uparrow$ accuracy) on the seismic dataset across all stimuli settings involving node-link diagrams, compared to power data.

H3: Both experts and non-experts prefer using cluster dendrograms over node-link diagrams. This hypothesis is fully supported. A Friedman test [Fri37] confirmed significant differences in preference across stimuli settings ($\chi^2(5) = 32.4$, $N = 45$, $p < 0.001$), with post-hoc Wilcoxon signed-rank tests [Wil92] confirming both dendrogram settings (C, F) were preferred over all node-link conditions ($p < 0.05$) (Fig. 6). We observed the following ranking across all participants for stimuli settings (from Fig. 2): (C) $1.03 \pm 0.08 > (F) 2.15 \pm 0.28 > (B) 3.54 \pm 0.42 > (E) 3.81 \pm 0.48 > (D) 5.34 \pm 0.28 > (A) 5.61 \pm 0.42$. Participants preferred (i) the box-glyph views for the clustered network and dendrogram, and (ii) tooltip reveal of FFT plots (D) over the underlying sensor network (A).

4.2. Study Implications

Event Tracking is Faster with a Tailored View. The performance with cluster dendrograms reflects how custom-designed visualizations can better support data-driven analytic needs, instead of relying on “standard” techniques, such as the node-link diagrams [PM09] that the baseline and composite views embodied.

Support for High Level Insights. ST2 consistently behaves as an outlier: the cluster dendrogram clearly visualizes flow contributions, giving participants $>90\%$ accuracy and high confidence, rendering the task trivial for factor analysis. ST3 was the most challenging task, yet showed the largest performance gains for power data and second highest for seismic data, indicating that cluster dendrograms particularly support high-level association tasks during epicentric analysis.

Design Implications for Non-Expert Users. Cluster dendrograms improved performance for both experts and non-experts, with non-experts performing comparably to domain experts for ST1 and ST2, indicating a low learning curve and broad accessibility for visualization non-experts [APSB23].

5. Conclusion

In conclusion, we evaluate cluster dendrograms via a controlled experiment wherein subjects perform spatiotemporal event tracking tasks using power transmission and seismic sensor data. Our results confirm that this approach streamlines epicentric event tracking across both domains, suggesting extensibility to broader time-varying sensor networks for higher-order tasks such as load balancing and early event flagging.

Supplemental Materials

All supplemental materials are available at https://osf.io/gtu7x/overview?view_only=a522595e057746de97fab46c9e6efca8, released under a CC BY 4.0 license. In particular, they include: (1) Excel files containing the aggregate data for collected measures, (2) Full set of stimuli used in the study along with source data visualizations, (3) Regression Analysis results (factor loadings, regression coefficients, significance), (4) demographic data of participants from Experiments, and (5) a full version of this paper with all appendices.

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